BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 038 · Operating Systems: Files and Devices

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What is an operating-system user interface for?

Q2. What does a denial-of-service attack try to do?
A. Improve network speed
B. Make a service unavailable
C. Create secure backups
D. Repair damaged hardware

Q3. What is data interception?

Q4. A login uses unsafe input in an SQL query. “SQL injection happens when a database has too many rows.” Correct this. How could an attacker affect the data?

Do Now Recall previous learning | **I Do** Read new knowledge | **We Do** Practise with guidance | **You Do** Work independently | **Review & Improve** Check my understanding

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

Peripheral — External or connected input/output hardware.

Device driver — Software enabling the OS to communicate with a particular device.

File management — Creating, naming, saving, organising and moving files/folders.

User management — Creating and controlling user accounts.

Access rights — Permissions controlling what a user may view or change.

Directory — A folder used to organise files and other folders.

New learning

Devices and drivers: The OS manages communication between the CPU and peripherals such as printers. A device driver provides software instructions that allow the OS to communicate with the specific hardware.

User management: The OS creates and manages accounts, passwords and permissions; staff and students may be allocated different access rights.

Files and folders: File management means naming, saving, copying, moving, renaming, deleting and organising files within folders/directories.

Choose the correct service: Device not responding? Consider its driver. A pupil cannot access staff data? Check account rights. Lost a saved document? Locate folders and file paths rather than blaming memory allocation.

Worked example

A school printer is installed and students can print from a PC. Explain the role of the operating system and a device driver. [4 marks]

1. Read the sequence: application → operating system → driver → printer.

2. The OS manages communication with peripherals [1] and processes a print request [1].

3. A printer driver provides device-specific instructions [1], allowing the computer to communicate correctly with the printer [1].

4. OS manages peripherals; driver enables communication with particular hardware.

We Do · Practise with guidance

Guided 1

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Q2. OS assigns different user accounts [1] and can authenticate the person signing in [1]. Access rights restrict guest permissions [1], so only authorised staff can edit booking records [1].

Q3. It names/saves/moves/organises files [1], allowing pupils to locate work [1]. It manages peripherals/drivers [1] so printers and other hardware can communicate [1]. It manages accounts [1] and access rights to restrict confidential files [1].

Exit Ticket · Model answer

True

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed:

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. A user interface lets people use the computer, through GUI controls or typed commands.

Q2. B — Make a service unavailable

Q3. Capturing data during transmission, potentially exposing confidential information.

Q4. Unsafe input can change an SQL query, letting an attacker view or change data. Explain how input changes the query and affects data access or privacy.

I Do / We Do pages · Guided practice

Q1. The OS manages communication with the projector [1] as an output peripheral [1]. A suitable driver enables software to send the commands in a format the projector can understand [1], allowing images to be displayed [1].

Q2. The OS uses user accounts [1] to identify the user [1]. Access rights/permissions limit file and folder access [1], preventing pupil accounts from changing payroll files [1].

Q3. It allows the pupil to name the file [1], save it to storage [1], create/name a folder [1] and move the document into the folder [1].

You Do pages · Code First

```
choice = 1
if choice == 1 then
    print("Save document")
else
    print("Move document")
endif
```

Expected result: Save document

You Do pages · Independent questions

Q1. Software that enables an operating system to communicate [1] with a specific hardware device/peripheral [1].

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

A college connects a new digital projector to a teacher's PC. Explain why a suitable driver and peripheral management are needed. [4 marks]

Worked method: The OS manages communication with the projector [1] as an output peripheral [1]. A suitable driver enables software to send the commands in a format the projector can understand [1], allowing images to be displayed [1].

Guided 2

A pupil account should not modify the school payroll folder. Explain how the OS helps protect files. [4 marks]

Worked method: The OS uses user accounts [1] to identify the user [1]. Access rights/permissions limit file and folder access [1], preventing pupil accounts from changing payroll files [1].

Guided 3

A pupil saves and moves a homework document into a new subject folder. Describe FOUR file-management operations the OS can provide. [4 marks]

Worked method: It allows the pupil to name the file [1], save it to storage [1], create/name a folder [1] and move the document into the folder [1].

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Code First · Predict → Write → Test → Improve

Predict the selected menu action.

```
choice = 1
if choice == 1 then
    print("Save document")
else
    print("Move document")
endif
```

predict

Question 1

State the purpose of a device driver. [2 marks] Source: Original OCR J277-style teaching question; not an official past-paper question.

Question 2

A hotel has guest and manager accounts on one shared computer. Explain how user management prevents guests from changing booking records. [4 marks] Source: Original OCR J277-style teaching question; not an official past-paper question.

Question 3

Explain the operating system's roles in file management, device management and user management for a school. [6 marks] Source: Original OCR J277-style teaching question; not an official past-paper question.

RHS COMPUTING / OCR J277 / 1.5.1 LESSON 038 · DRAFT

BLACK PEN Answer all questions first. GREEN PEN Then: tick correct answers; rewrite mistakes.

Exit Ticket · Answer before checking

A device driver helps the operating system communicate with particular hardware.